

Where the project stands

Goal: rewrite NASA GISS's ROCKE-3D (Fortran GCM) in Python/JAX, checked against the real Fortran, for GPU use. Configuration P2SAoM40 (72x46 grid, 40 layers, dynamic ocean). Every card says what is met and what is not.

NOT MET

One-day criterion (nov26, 54 steps)

ACCEPTANCE s9: never beyond 2x the largest real-member distance at every step. QCL at step 1 is 2.84x (one bistable cell, D197). Steps ≥ 3 : worst ratio 1.69 (QCI, inside the run-to-run spread). Two days run. D197, D207, D213.

MET / PARTLY MET

One coupled step vs real Fortran (C2, step 0)

nov26 MET with one named column (25,16); dec01 and jan01 PARTLY MET (worst EGCM 9.9e-9 and 7.6e-9 of scale). Needs the libimf callback. D191.

18.8 s

Steady step on CPU, was 44.3 s

99 jit executions (was 262), eager dispatches 490 (was ~15,700). Only 1.2-1.4x faster than the NumPy chain on the same cores. HYBRID step. D187, D191.

558 / 558

C1: bitwise vs our NumPy chain

Step 0 on nov26, dec01, jan01 (category A). On the day: bitwise through step 47 with the computed GHY schedule; the NumPy chain has no daily_LAKE. D191, D204, D206.

3,611 passed

Last batch regression (15baa26)

6 skipped, 12 failed: all test_ghy_jax cases on the new day-2 records (abetad beyond iteration 11); that directory is now excluded, no tolerance changed; the file re-run alone: 219 passed. README_START_HERE handoff.

No GPU result

GPU runner built, validated on CPU only

A100 probe: launch-bound for small sequential ops; device exp/sin/pow differ from NumPy by 1 ulp in 6-12% of values, so never bitwise on GPU. D198, gpu/DISCOVER_RUN.md.

This is a HYBRID port, not end-to-end JAX: radiation is the original Fortran (served) or replayed from a record; libimf math, QUS, pole columns and the OADVT2 pre-pass are host callbacks; Ent exports, land forcing and tile radiation columns are recorded inputs (D191 s4). SOCRATES is never ported or modified.

Status by area, and what is still outstanding

Green = verified success, yellow = in progress or partial, red = broken or not met, grey = not started. Violet marks outstanding items and their next action.

Status by area (2026-10-08)

- Component ports vs the Fortran (Track B)**
validated rung by rung, D4-D145
- One coupled step vs the real Fortran (C2)**
step 0 MET on nov26, dec01, jan01 (D199); hybrid
- JAX step equals the NumPy chain (C1)**
bitwise: 561/561 at step 0; nov26 steps 0-47 (D206)
- Regression suite**
3,611 passed, 6 skipped; 12 day-2 test cases excluded, fixed file 219 passed (15baa26)
- Device-resident assembled step**
99 jit executions, 18.8 s/step; hybrid, not end-to-end JAX
- Radiation (SOCRATES)**
Fortran server callback works (D210/D212); not ported
- GPU assembled step**
runs on the A100 (step 0 OK) but NaN from step 1 (surface stage); cause unknown
- Second model day (steps 54-107)**
real records made (D207); day 2 never beyond 2x of the members
- One model day vs 5 real members (ACCEPTANCE s9)**
NOT MET: QCL step 1 = 2.83x in every run (systematic); rest inside the spread (D213)
- Daily lake update at the day boundary**
ported, bitwise vs the compiled Fortran (D205, D208)
- Months-long run and its criterion**
not started; criterion not defined

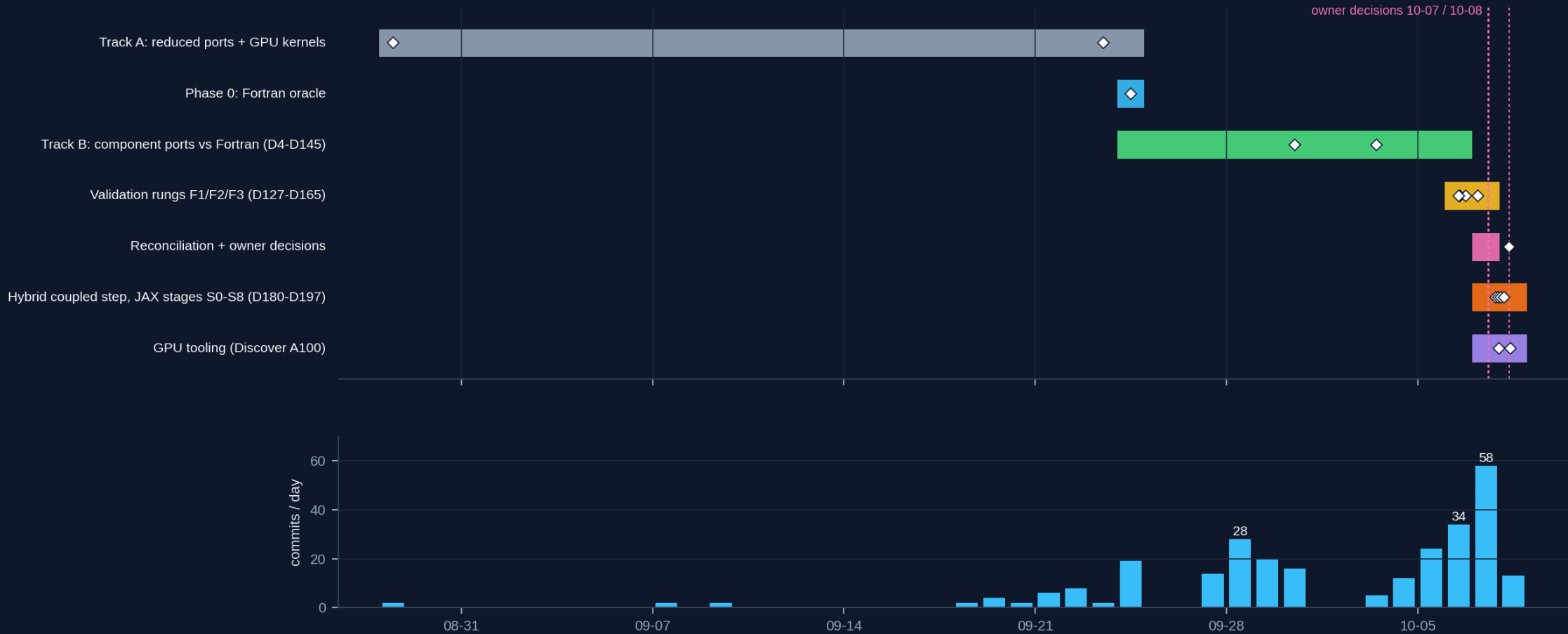
Outstanding items and their next action

- GPU NaN at step 1 (surface stage)**
owner: JAX_DEBUG_NANS=1 run on Discover; CPU control with the simplifier on is finite
- Replayed inputs in the step**
Ent exports, land forcing, tile radiation columns, PBL templates (D212 list); tile radiation columns next
- QCL step 1 exceedance (2.84x)**
one bistable cell (D197); owner may relax s9 later (dated, both statuses)
- Free-running comparison**
compute Ent and land forcing instead of replaying the real records
- Radiation and SOCRATES**
135-295 h port estimate; only if the owner lifts the rule (D1-D6)
- Multi-month criterion**
define it; use the 8-member JAN1950 leave-one-out scoring
- NumPy reference beyond day 1**
give the NumPy chain a daily_LAKE so C1 holds across the boundary
- Housekeeping**
owner: close extra sessions; decide where ANN4099 lives; push when batches pass

Honest status: step 0 matches the real Fortran on three dates and two model days run, but the step is a hybrid with replayed inputs and the one-day statistical test is not passed (QCL step 1).

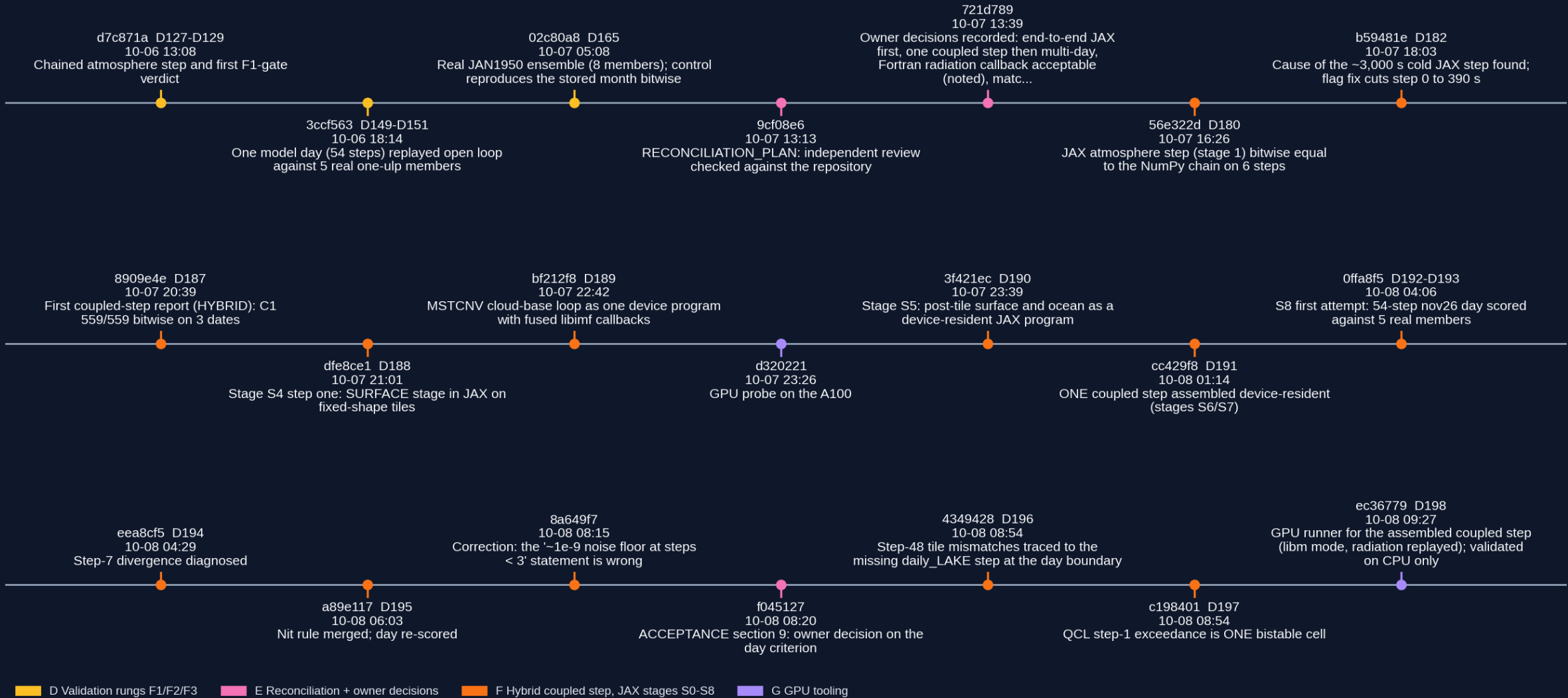
Six weeks, 273 commits: from reduced ports to a hybrid JAX coupled step

ROCKE-3D -> JAX: chronology, 2026-08-28 to 2026-10-08 (273 commits; diamonds = milestones listed in PROJECT_CHRONOLOGY.md)



From the F1 gate to the one-day verdict: 2026-10-06 to 2026-10-08

Zoom: 2026-10-06 to 2026-10-08, events in commit order (not to time scale); date/time = commit time; colors = phase



Seven phases, seven lessons

Phase	Dates	Key commits / ledger	What was learned
A Track A: reduced component ports, GPU kernels	08-28 to 09-24	c2991dd, 5cbe7f8, abd805b	Reduced drivers ran fast (kernels 2.3-6.3x on GPU; fused chain 4.2x) but were not faithful; Track A recommended against a full port, later reversed by the owner.
B Phase 0: Fortran oracle	09-24	97e2457, 9cd4bc6, 4ec6056 (D3)	Instrumented real ModelE reproduces the restart bitwise. Chaos floor: 1-ulp perturbation reaches ~1-3% of variability in 5 days, so acceptance beyond one step must be statistical.
C Track B: components vs real Fortran	09-24 to 10-06	D4-D145; D29, D36, D51, D55, D121-D123	Dump-hook-and-validate found real bugs (3 in DYNSI, OMEGA constant, 2 land errors D135/D136) and dead code (OCNDYN.f legacy driver, OCNTDMIX.f, OCNQUS.f).
D Validation rungs F1 / F2 / F3	10-06 to 10-07	D127-D129, D149-D157, D163, D165	Single-step gate needs libimf (3-5% of cloud columns flip without it). One-day runs sit inside the real model's own spread except condensate tails. Radiation can be served by real RADIA (D152-D162).
E Reconciliation + owner decisions	10-07	9cf08e6, 721d789, b71eb22, bd9b16c	Fidelity work was ahead of the JAX deliverable; criteria were fixed BEFORE the coupled step was compared; C1 and C2 reported separately.
F Hybrid coupled step, stages S0-S8	10-07 to 10-08	D180-D197	Assembled device-resident step bitwise to NumPy (C1), C2 at step 0 MET/PARTLY MET; the 54-step day is NOT MET; every day diagnosis found a cause, none a port defect in clouds.
G GPU tooling	10-07 to 10-08	5c5b7a4, c7c42c8, d320221, ec36779 (D198)	Environment works on an A100; small sequential ops are launch-bound; no bitwise on GPU; no GPU run of the step yet.

Categories A / B / C / D, the gate, and two comparisons

Per prognostic field, maximum absolute difference as a fraction of the field's scale (ACCEPTANCE s3; categories fixed by the project plan and not loosened).

A bitwise equal	B $\leq 1e-12$ of scale (rounding level)	C $\leq 1e-6$ of scale	D worse than C
------------------------	---	-------------------------------	-----------------------

Verdict for one coupled step

- MET: every prognostic field in A or B, except named exception columns (at most 10 per field, each $\leq 1e-9$ of scale).
- PARTLY MET: runs end to end with named exceptions in B or C and the first failing stage identified.
- NOT MET: otherwise. Reported per date (nov26, dec01, jan01), never pooled.
- Bitwise (A) needs the Intel libimf runtime; without it results are B or C and say so.

C1 versus C2 (reported separately)

- C1, port consistency: JAX-driven step vs our NumPy chained step, same start state, cores, flags, no compile cache. A pass says JAX reproduces our NumPy port, NOT that it reproduces ROCKE-3D.
- C2, fidelity: JAX-driven step vs the real Fortran step-boundary dumps. Only C2 supports a statement about ROCKE-3D.
- Hazards D139, D145, D175: results differ across core counts and with a warm compile cache.

Multi-day rung (F2, statistical), ACCEPTANCE s4 and s9

- Reference spread: five real one-ulp members of the nov26 day (D151). Classes per field and step: within ($\leq$ largest member distance), near ($\leq 2x$), beyond ($> 2x$).
- Owner decision 2026-10-08 (s9): the pass rule is 'never beyond 2x the largest member distance, at every step, for every scored field' over ALL steps; 'within at every step' is a secondary line.
- Longer windows: the D172 leave-one-out scoring with the 8 JAN1950 members (D165); the earlier ' $\geq 95\%$ of zonal bins within 2 sigma' rule is not used (true members fail it).

Identical inputs: what 'the same run' means, and what is compared at each horizon

Same restart, boundary data and configuration (P2SAoM40) for the Fortran and the JAX port; how exact the match can be depends on how far you run.

Horizon	What is compared	What 'match' means	Status (2026-10-08)
One step	Full model state after the step: T, U, V, Q, P, cloud water and ice, moment arrays; ocean, ice, lake and land state, field by field	Categories A bitwise, B $\leq 1e-12$ of scale, C $\leq 1e-6$, D worse; met if every gate field is A or B except named columns	nov26, dec01, jan01 MET at step 0 after D199 (nov26 with named column (25,16))
One model day (54 steps)	The same fields at every step, as distance from the real run	Statistical: distance $\leq 2x$ the largest spread among five real one-ulp members at every step (ACCEPTANCE s9); bitwise is not expected (chaos)	nov26 NOT MET (D195, QCL step 1 = 2.84); rerun on the D199 land code in progress (D200)
Months and longer	Climate statistics: zonal means, variability, energy balance	Not defined yet; 8-member JAN1950 ensemble and D172 leave-one-out scoring exist, no long run exists	not started

Identical inputs: what that includes today

- Same restart state, boundary data, orbital and configuration settings, same physics code version: our tests start from the real model's own state at a step boundary.
- The model is chaotic: two correct runs drift apart, so beyond a few steps agreement can only be statistical.
- Bitwise matching needs the Intel libimf math library (host callback); on the GPU even exp/pow differ by 1 ulp in 6-12% of values.

Not yet an independent same-parameters run

- Recorded inputs: vegetation (Ent) exports, land forcing and some surface columns are taken from the real model's records each step, so the run is not free-running.
- Radiation is replayed from the record or called through the original Fortran; SOCRATES is not ported.
- The daily lake update at the day boundary is not ported (D196). A real blind comparison needs these computed, and a multi-month criterion defined.

Single-step (step 0) verdicts over time: honest status per date

dec01 and jan01 were PARTLY MET in every ported-land configuration until D199 fixed the land humidity (step 0, one run each); nothing here is claimed beyond what the ledger says.

Date / entry	Configuration	nov26	dec01	jan01	Note
10-06 D129	libimf, RECORDED land patch	MET (named cols)	MET (A/B)	MET (named col)	nov26 W2GCM 4 cols 1.7e-12; jan01 1 col <= 2.5e-12
10-06 D129	libimf, ported GHY	NOT MET	PARTLY MET	PARTLY MET	first failing stage: SURFACE/GHY land cell (62,34)
10-06 D129	no libimf (libm)	NOT MET	NOT MET	NOT MET	first failing stage CONDSE: cloud threshold flips in 3-5% of columns
10-06 0ad795b	libimf, ported GHY after D135/D136 fixes	MET	PARTLY MET	PARTLY MET	all MET with recorded land
10-07 D187	HYBRID assembled (libimf callback, replay radiation)	MET, 1 col	PARTLY MET	PARTLY MET	fails ACCEPTANCE items 1 and 3; 44.3 s/step
10-08 D191	assembled device-resident step	MET, 1 col (25,16)	PARTLY MET	PARTLY MET	worst EGCM 9.9e-9 (dec01), 7.6e-9 (jan01); item 1 holds only under owner decision 8.3 with named exceptions
10-08 D199	assembled step after the land qg_aver/elhx fix	MET, 1 col (25,16)	MET (1/13/0/0)	MET (1/13/0/0)	worst gate field 4.1e-13 (dec01), 6.3e-13 (jan01); C1 559/559; step 0 only, replayed radiation, libimf callback
10-08 D195 / s9	54-step nov26 day vs 5 real members	NOT MET	not run	not run	QCL step 1 ratio 2.84; steps >= 3 worst 1.13
10-08 D198	libm-mode GPU runner, CPU check	n/a (not fidelity)	-	-	vs NumPy libm: 555 A + 3 B of 558; vs libimf CPU: A126 B170 C47 D215 (a property of libm, not of the port)

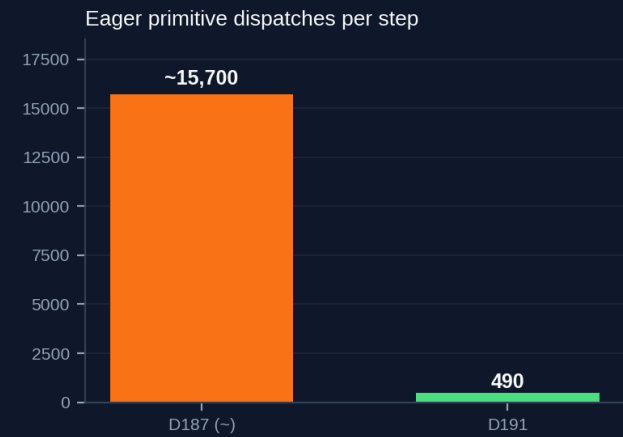
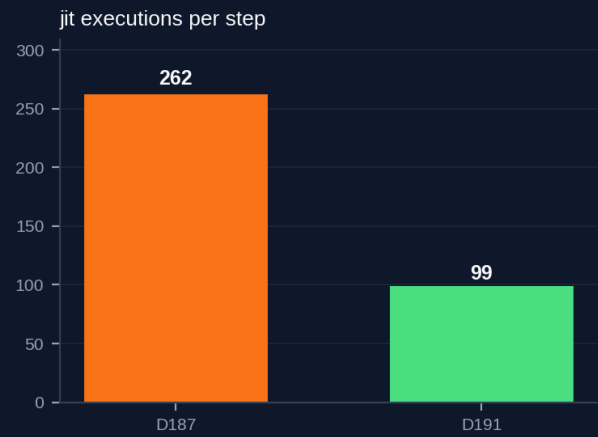
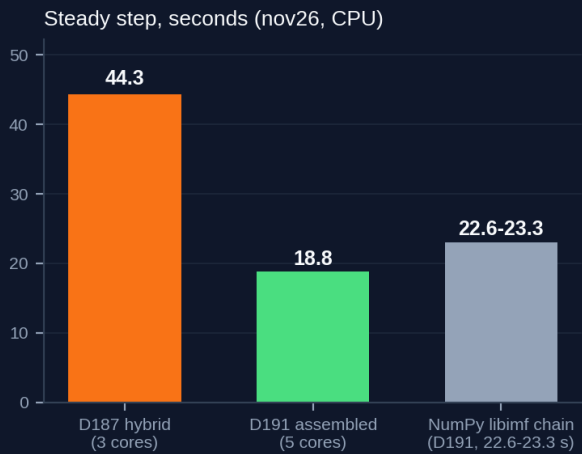
dec01 / jan01 were PARTLY MET because of one land humidity term (D199, fixed at step 0 only; later steps and the day are being re-measured). Four surface-composite exports (QGAVG, USAVG, VSAVG, TGVAVG) were category D against the real end record in D187 and were not re-measured in D191.

The assembled coupled step (D191) is a hybrid

Component	How it runs in the assembled step	Status	Source
Atmosphere phase 1 dynamics (J1)	JAX; 75 kernel dispatches from Python; pow through libimf host callback	JAX + callback	D186, D191
Clouds / convection	LSCOND and MSTCNV on device (D189, fused libimf callbacks); QUS subsidence (176 calls, 249.5 MB) and 2 pole columns are host callbacks	JAX + callbacks	D189, D191 s4
Radiation (SOCRATES)	Original Fortran RADIA via persistent server (~11 s/call, every 5th step, 33.7 MB to host, 11.6 MB back) or REPLAYED from the real record. Never ported.	Fortran / replay	D159-D162, D185, SOCRATES_PORT_PLAN
libimf pow/exp	Host callback into the original build's Intel runtime (the fidelity configuration); GPU/libm mode has none	host callback	ACCEPTANCE s8.1, D183
Surface tiles + land (GHY)	JAX SURFACE stage on fixed-shape tiles (D188); Ent exports, land forcing and tile radiation columns are RECORDED each step	JAX + recorded	D188, D191 s1
Post-tile surface + ocean	Device-resident JAX (GROUND_*, RIVERF, DYNSI, OCEANS, FORM_SI, ADVSI); OADVT2 east-west pre-pass is a host callback	JAX + callback	D190, D191
Phase 2 (DISSIP, FILTER)	jnp, bitwise vs NumPy; pow of SLP/MAtPMB/PEK through libimf callback	JAX + callback	D191 s1
Host work every step	template_build (eager, 294 dispatches), record load (200 calls, 138 MB), day boundary DAILY_ATMDYN in NumPy	host	D191 s4, D195
Not implemented	new-ice-tile donor rule; daily_LAKE (D196); daily ocean/ice/land updates; transfer-guard test of the full step	missing	D191 s1, D195 s6, D196

D191 verdict on ACCEPTANCE s1: items 2-5 hold as reporting items; item 1 holds under owner decision 8.3 (three jit units plus host calls) for the prognostic state, with named exceptions. 'I do not call it end-to-end JAX.'

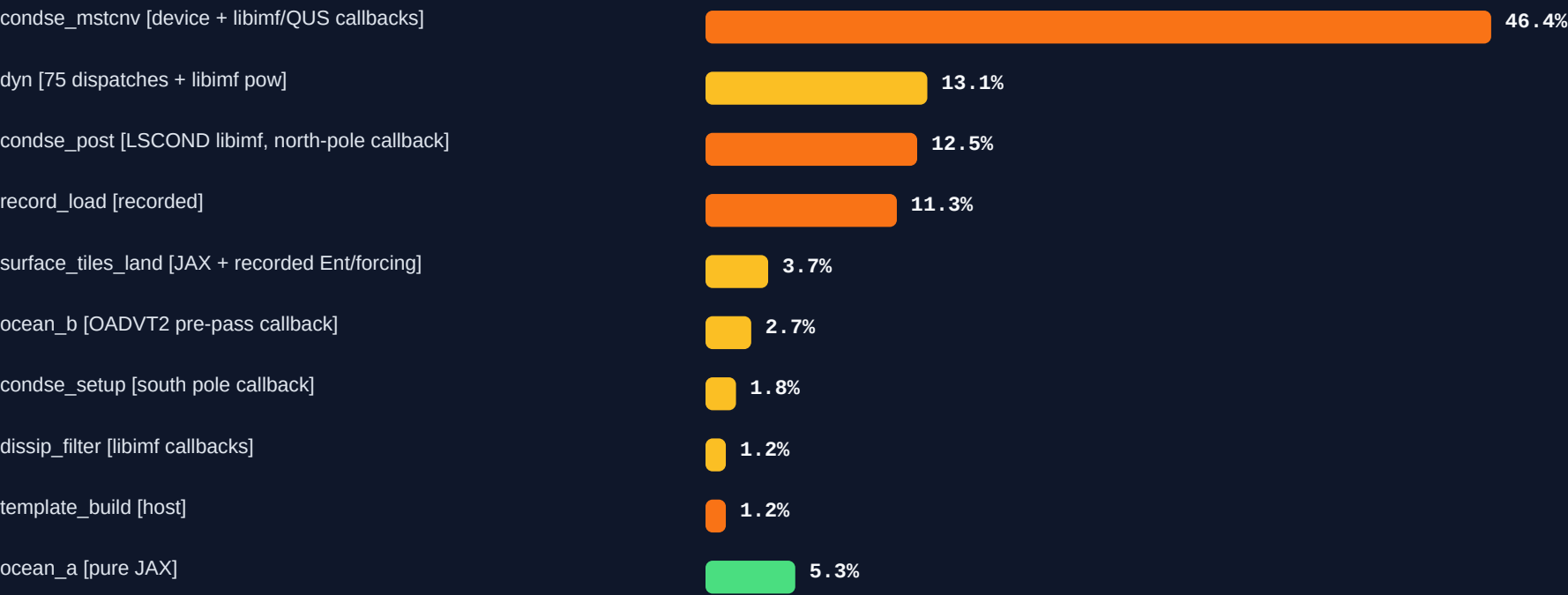
44.3 s to 18.8 s per step; 262 to 99 jit executions



Entry	Change	Before -> after	Note
D182	cold-compile pathology: XLA flags algsimp + reshape-mover disabled	~3,000 s -> 390 s cold step 0 (commit 7131447: ~30x faster cold compile)	results bitwise identical
D188	SURFACE stage in JAX on fixed-shape tiles	~2x faster, zero eager dispatches inside	category A on 3 dates
D189	MSTCNV as one device program, fused libimf callbacks	phase 1 of step 0: 33.6 s -> 11.9 s	bitwise equal
D190	post-tile surface + ocean as JAX	123 -> 5 jit executions; 8.0 s -> 1.76 s	149/149 category A
D191	assembly of J1 + J2 + J3	steady 44.3 -> 18.8 s; cold first step 349 -> 490 s (worse)	shared node, one run per number
D191	vs NumPy libimf chain, same cores	only 1.2-1.4x faster (22.6-23.3 s)	46% of step is MSTCNV + callbacks, 13% dynamics dispatch
D198	libm-mode runner on CPU	cold 572 s, steady 11.0 s (load average ~20)	indicative; GPU time unknown

Callbacks and recorded-input work still dominate the assembled step

D191 section 4, nov26, steady, timed run with a device block after every stage (step 17.8 s, stage sum 20.1 s incl. record load).



- Callback bodies (libimf_ops 5.0 s, fused 0.68 s, QUS 0.78 s, poles 0.39 s, OADVT2 0.26 s = 7.1 s) are included in these shares: about 40% of the step. Radiation: 0.04 s replayed; 12.7 s per call with the real server (1 call per 5 steps).
- Purely-JAX stages (post_a, post_b, surface_pre, template_apply, tile_acc, carry_writeback, melt_si_dev, state_assembly) are each <= 0.2%. The kind labels (JJ/NP/REC) were set by hand from code inspection; the shares are measured.
- Over the 54-step day (D195): steady median 15.9 s (min 15.1, max 18.7); 13 SURFACE rebuild steps of 68-74 s each; cold step 0 428 s; total wall 2,128 s.

Which differences between our runs mean something (D207, D209-D213)

Run (54 steps, nov26)	T	U	V	Q	P	QCL	QCI	worst ratio steps >= 3
D209 replayed radiation (default)	54/0/0	54/0/0	54/0/0	42/12/0	52/2/0	52/1/1	37/17/0	QCI 1.69
D210 server radiation (Fortran callback)	54/0/0	54/0/0	53/1/0	54/0/0	52/2/0	52/1/1	49/5/0	QCI 1.08
D212 server + own surface fields, seed, COSZ	54/0/0	53/1/0	54/0/0	54/0/0	51/3/0	41/12/1	42/12/0	QCL 1.41
D213 spread (control, p1, p2, p3), within min..max	53..54	53..54	54	42..49	52..54	50..52	37..52	QCI 1.07..1.84
Day 2 (steps 54-107), replayed radiation	54/0/0	54/0/0	54/0/0	17/37/0	54/0/0	53/1/0	52/2/0	QCI 1.10

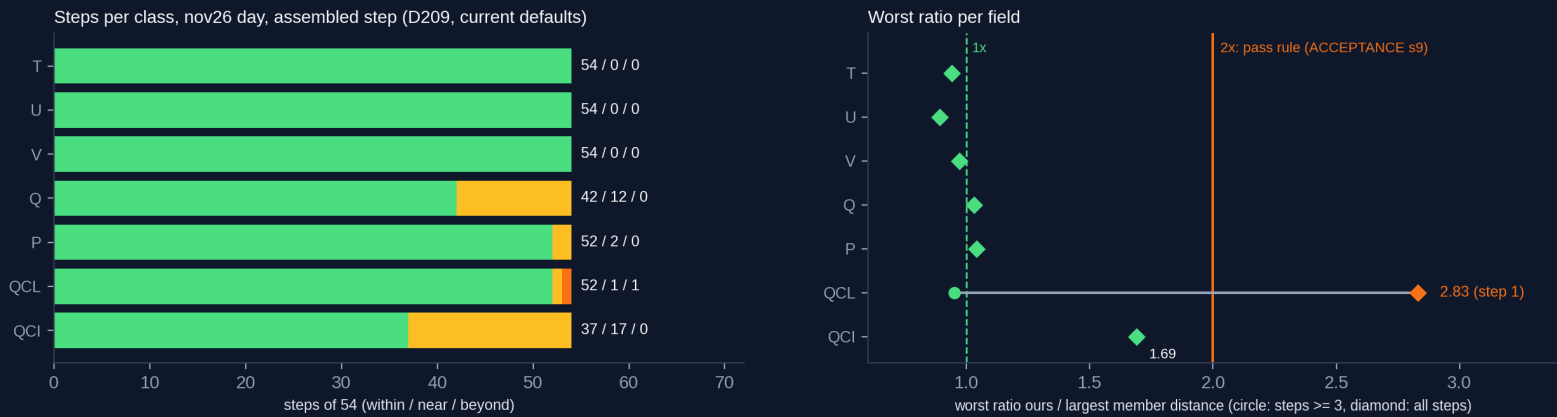
What the spread says

- One-ulp perturbations like the real members' move our QCI within-count from 37 to 52 and Q from 42 to 49: the D209 vs D210 differences are inside that spread.
- QCL at step 1 is 2.83..2.92 in EVERY run and configuration: systematic, not noise.
- D210/D212 Q = 54 and D212 QCL = 41/12/1 lie outside a 3-run range, which underestimates the spread: single unconfirmed results.

Conditions and limits

- HYBRID. Ent exports, land forcing, tile radiation columns and PBL templates are replayed from the real records; radiation is replayed, or computed by the unmodified Fortran server (D210, D212).
- Day 2 reuses the same member runs (perturbed at step 0): its floor is a continuation, not a fresh perturbation. Replayed records pull both days toward the real trajectory.
- ACCEPTANCE s9 stays NOT MET (QCL step 1). Tables are single runs; n = 3-4 realisations.

One model day (nov26) vs five real members: NOT MET under ACCEPTANCE s9



Field	within / near / beyond D209 (D203)	worst ratio steps >= 3	worst ratio all steps	near steps	beyond
T	54/0/0 (54/0/0)	0.94 (step 15)	0.94	-	-
U	54/0/0 (53/1/0)	0.89 (21)	0.89	-	-
V	54/0/0 (54/0/0)	0.97 (10)	0.97	-	-
Q	42/12/0 (40/14/0)	1.03 (46)	1.03	40-53 (12)	-
P	52/2/0 (54/0/0)	1.04 (33)	1.04	2	-
QCL	52/1/1 (50/3/1)	0.95 (9)	2.83 (step 1)	1	step 1
QCI	37/17/0 (49/5/0)	1.69 (47)	1.69	17 steps	-

Status: NOT MET (owner decision 2026-10-08, ACCEPTANCE s9). The only field beyond 2x is QCL at step 1 (ours 1.15e-6 vs largest member distance 4.06e-7). All 54 steps are finite since the D202 land fix. QCL step 1 (2.83) is systematic in every run; the other fields vary within the run-to-run spread (next slide). 'Within at every step' also fails (Q, P, QCL, QCI near at some steps). The earlier claim of a ~1e-9 floor at steps < 3 was wrong (correction 8a649f7); no exception clause is recorded.

Conditions: HYBRID; radiation REPLAYED (not computed); libimf host callback; Ent, land forcing, template columns recorded; ocean/ice/land daily updates not applied; nit_strict=False after step 22; one start state; five members. D203 (before the computed GHY schedule, D204) in brackets.

D193-D197: each exceedance traced to a cause

D194 -> D195 (fixed)

Step 7 divergence

Not the tile set. GHY cells with $\text{ffnit} > 11$ must regenerate sub-iteration lengths from OUR precipitation; the step used the recorded one. Fix merged (nit_fix default on): C1 bitwise steps 0-22. The NumPy reference aborts at step 23 (cell (41,21): 11 vs 14 iterations). D193's 'rebuild when the tile set changes' was wrong: the cause is `max_substeps`.

D196 (open, patch not applied)

Step-48 tile mismatches

54 lake cells ($\text{FOCEAN} = 0$) lose their open-water tile because `daily_LAKE` (LAKES.f:2492) is not applied at the day boundary (FLAKE changes in 632 of 636 lake cells at 47->48). Our NumPy chain has the same omission. Level-2 fix is a ~500-line port, not started.

D197 (explained, still NOT MET)

QCL step 1, ratio 2.84

One bistable cell (31,12,15) holds 96.8% of the squared error. 1-ulp PK differences from the dynamics exit flip a stratiform threshold; 1 of 8 random 1-ulp draws reproduces it exactly. CLOUDS port from real inputs is category A/B; the NumPy chain has the same value. Without that cell the ratio is 0.51. Does not relax the criterion.

D199 -> D203 (fixed)

Land runaway, NaN at step 39

D199 (land humidity uses the current substep's `elhx`) fixed `dec01/jan01` at step 0 but sent the `nov26` day to NaN (cell 41,20). D201: the `elhx` switch alone decides finite vs NaN. D202: GHY must receive the PBL's `gusti` (GHY_DRV.f:1267), not the recorded one. After that all 54 steps are finite; QCL step 1 (2.84x) is unchanged. Mechanism of the original trajectory change is not proven. Correction 8a649f7 (noise floor) stays in force.

The real model is chaotic; our step is judged against its own spread

Noise floors measured with the real binary

- D3 (5 days, 1-ulp T perturbation): T layer-1 pointwise rms 0.048 K, global-mean 1.4e-4 K; whole column T rms 0.27 K; only 36 of 257 restart variables stay bitwise identical.
- D151/D149: five real members for the nov26 day; our NumPy and JAX runs diverge from each other exactly like real members.
- D165: 8 real JAN1950 members (ctrl + 7 one-ulp); ctrl reproduces the stored month bitwise; t_{500} global-mean sd 0.043 K; 8 members = about +/-25% on a std; one season.

Earlier day results (replayed / free radiation)

- D151 open loop: T, U, V within at all 54 steps; Q near 15 steps, condensates near ($\leq 1.11x$); none beyond $2x$.
- D155-D157 free-running radiation (real RADIA server): T/U/V within at every step; condensates up to $1.95x$ (QCI step 47); radiative flux differences < 1 W/m² global mean.
- D171 (Ent computed): QCI worst 1.62; QCL step 1 ratio 2.90 (D192 re-score). The D192 scorer reproduces D157 and D171 exactly.
- None of these days is 'within at every step' for all fields; they were reported before the 2026-10-08 pass rule.

Consequences

- Single steps are deterministic (floor 0) and can be bitwise only with libimf; beyond step ~1 any rounding difference grows, so acceptance is statistical (ACCEPTANCE s4).
- Without libimf, cloud thresholds flip in 3-5% of columns: a multi-step run drifts about 0.1-0.2 K after 6 steps (GOAL.md; D129).
- Limits: five members, one start state, one day; heavy-tailed condensates (QCI, QCL) are single-cell dominated, so a ratio near 2 is not a significance statement (D192).

What can silently change a bitwise or statistical result

Hazard	What was found	Source
Intel libimf vs glibc pow/exp	Bitwise needs the real build's libimf. With NumPy/glibc dynamics differ at ~1e-13 and cloud columns flip in 3-5%. pow/exp/log are not correctly rounded: ~0.02% of a KPP table differs by 1 ulp (D54).	README caveats; D54; D129
Single-precision literals	Fortran literals without d0 (e.g. <code>**(1./3.)</code> in OCNKPP.f) are single precision under the build; <code>REAL*4 DZDH1</code> in RIVERF.	D54, D167
REAL*16 sites	Only Ti/Ti2b in SEAICE.f; a binary128 emulation makes seaice_to_atmgrid and ADDICE bitwise; not applied to existing modules.	D173
XLA CPU flags	FMA fusion and x/c rewrites break bitwise; needs <code>--xla_cpu_max_isa=AVX</code> and algsimp (and reshape-mover, D182) disabled before importing JAX; such tests need separate processes.	README caveats; D182
Core count / compile cache	Existing code gives different results on 1 and 3 cores; a warm JAX compile cache changes results. Use identical affinity and no cache for validated runs.	D139, D145, D175
Host callbacks on one core	<code>io_callback/pure_callback</code> inside jit DEADLOCK on a single core (0.1% CPU for 23 min). Give such jobs <code>>= 2</code> cores.	D185; README
NumPy vs jnp semantics	<code>x**4</code> is libm pow in NumPy but repeated multiplication in jnp; <code>jnp.cumsum</code> is not a left-to-right sum.	SOCRATES_PORT_PLAN s3
GPU device math	<code>exp</code> , <code>sin</code> , <code>pow</code> differ from NumPy by 1 ulp in 6.14%, 11.96%, 9.07% of values (log 0.22%): category A is unattainable on GPU unless every call goes through a host callback.	gpu/DISCOVER_RUN.md

The A100 probe says: fuse into few large kernels, and do not expect bitwise

Measured on Discover (NVIDIA A100-SXM4-40GB, JAX 0.6.1 CUDA)

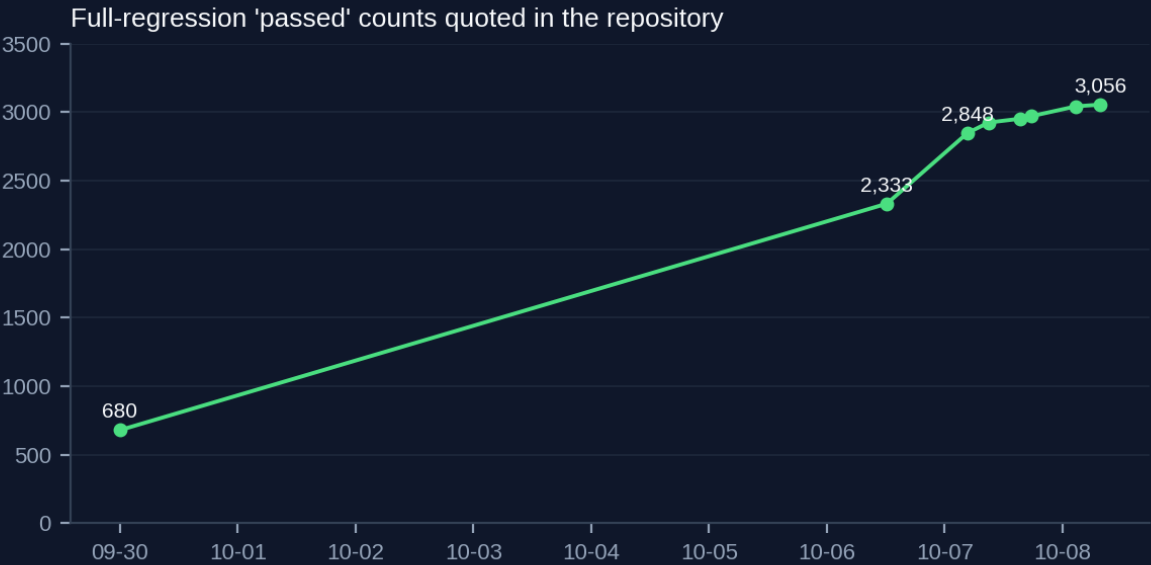
- Smoke test 2026-10-07 (job 58776063): 27 pass, 2 warn (no pytest, no libimf), 0 fail; float64 works; jit+scan on model-sized state compiled in 0.6 s.
- Probe (job 58778982): float64 matmul 2048x2048 1.2 ms (about 820 ms on one CPU core of the work node).
- 100,000 sequential scan steps: 575 ms on the A100 vs 41 ms on one CPU core: launch-bound, ~5.7 us/step; the probe note calls it 14x slower. Long chains of small ops lose; only wide parallel work wins.
- Device exp 6.14%, sin 11.96%, pow(x,0.25) 9.07%, log 0.22% of values differ from NumPy (max 1 ulp).

What D198 built, and what the first GPU runs show

- gpu_step_run.py runs the ASSEMBLED step in libm mode (no libimf), radiation replayed, no NumPy reference needed; data subset 6b (66 files, 1.58 GB), runbook MODE=step.
- On CPU: libm step equals the NumPy libm chain (555 A + 3 B of 558). Against the libimf CPU result: A126 B170 C47 D215: threshold fields flip with ANY different arithmetic (another XLA version: 198 of 558 category D; no CPU flags: 208).
- RUN on the A100 (jobs 58786705, 58788593, 58789387): step 0 finite, within the CPU libimf-vs-libm spread (T 7e-7, Q 2e-4). Cold 726-768 s (655-689 s compile). Steps >= 1: NaN from the surface stage (land/ice/lake/ocean); atmosphere phase 1 is clean; error flags did not catch it.
- A CPU control with the simplifier on is finite; cause on the GPU not known yet (JAX_DEBUG_NANS run pending). The 13.1 s step was a repeat of step 0; steps 2-5 took 25 s (NaN steps), the CPU 10-19 s.
- 99 jit executions, ~490 eager dispatches, 176 QUS callbacks per step: launch-bound; the step is not faster on the GPU so far.

Honest label for any future GPU result: libm mode, radiation replayed, recorded inputs, HYBRID, not bitwise to CPU or Fortran; judged at rounding-level categories for one step and statistically beyond. It must not be used for a C2 / fidelity claim (D198 parent check).

3,611 passed, 6 skipped at 15baa26 (12 day-2 cases of one test file excluded afterwards)



- Batch regression of 15baa26 (clean git-archive export, JOBS=4, cores 8-11, 4,652 s): 3,611 passed, 6 skipped, 12 failed (all test_ghy_jax on the nov26_day2 records; excluded afterwards, file re-run alone: 219 passed). The earlier export artifact (test_jax_harness) is fixed.
- The failure, test_jax_harness::test_header_complete, asserts a 40-character git HEAD that an export lacks; passes in the real tree. The test was then changed to tolerate a missing .git (commit 1036056); the full suite was NOT re-run after it.
- The 6 skips are gated tests (environment switches) not run in the batch.
- Mutation and non-vacuity checks accompany most ports; flag-sensitive JAX tests run in separate processes (run_all_tests.sh; sharded runner 34 min vs ~83 min serial, aggregate totals equal).
- Rule of the owner (2026-10-07): large batches; no full-suite re-run per code change.

Passed counts are what the repository records at each commit (680 after D51 per Project_Summary; 2,333; 2,848; 2,925; 2,953; 2,972; 3,043; 3,056). They count tests, not validated physics: acceptance is the ledger's per-field categories and the scorer of D192. The 2,471 + 206 entry of 264283d is left out (ambiguous).

Open work, grouped by what it blocks

Blocks the day criterion

- Day boundary: apply daily_LAKE (D196, ~500-line level-2 port, not started); other daily updates (ocean/ice/land) not applied.
- New-ice-tile donor rule (D190 s6): not implemented; multi-day runs without per-step records need it.
- NumPy reference aborts at step 23: C1 beyond step 22 unavailable; nit_strict policy beyond step 22.
- Leave-one-out scoring with the 8 JAN1950 members (D172) for longer windows: not implemented for the coupled step.

Blocks 'end-to-end JAX'

- Radiation stays Fortran (served/replayed): SOCRATES port is planning only, 34,899 reached lines, 135-295 h estimate, rule unchanged.
- Recorded inputs: Ent exports (stage-1 port exists, not in the assembled step), land forcing, tile radiation columns, CONDSE entry set, PBL profile columns; AG2OG/IG2OG and straits start in ocean chain.
- Host callbacks (libimf, QUS, poles, OADVT2) and host template build; no transfer-guard test of the full step.
- dec01/jan01 PARTLY MET from the ported land surface (not re-diagnosed).

Blocks the project's purpose (GPU)

- First GPU run of the assembled step (D198 runner ready; MODE=step).
- Fusing the step into few large kernels (probe: launch-bound).
- Speed without the libimf callback was not measured (D191 s5).
- Reference identity: our run is not shown to be the supplement run P2SAoM40_003 (PROVENANCE_MANIFEST); the 100-year climatology comparison is a later milestone, not tested.

Effort: no current total is derived. GOAL.md (10-06) says ~100-170 h more to a month comparison; README_START_HERE (10-07) says that figure is too high but gives no new total; treat any total as provisional.

Owner decisions taken (10-07, 10-08) and decisions still pending

Taken (recorded in the repository)

- 2026-10-07 (GOAL.md): 'port' = end-to-end JAX first, validated components second; success = one coupled step first, then multi-day; a Fortran radiation callback is acceptable but must be labelled; match P2SAoM40 first; one-month and 100-year comparisons are later milestones.
- 2026-10-07: ACCEPTANCE_CRITERIA approved before any coupled step was compared; later changes must be dated, never silent.
- 2026-10-07 (s8): headline C2 uses a labelled libimf host callback and is also reported in libm mode; three jit units plus host calls satisfy s1 if boundaries and non-JAX stages are reported; D176 files adopted, D178 after verification.
- 2026-10-07: Ent exports stay recorded up to one day; Ent port started. One writer per working tree. Test/push rule: large batches.
- 2026-10-08 (s9): day pass rule = never beyond 2x at every step; nov26 day NOT MET; no exception for steps 0-2; relaxation only later, only to advance to the GPU run, dated, with both statuses.

Pending for the owner (as the documents stand)

- Whether to relax the day criterion to advance to the GPU run (s9.3): must be a dated decision after the results, stating the unrelaxed status next to it.
- nit_strict policy beyond step 22 and what to do about the NumPy reference abort at step 23 (D195).
- Whether to implement daily_LAKE (D196 patch proposal, level 2 ~500 lines) and the new-ice-tile donor rule.
- Name the GPU host formally (ACCEPTANCE s7.4 / s8.5 still list it open; Discover A100 smoke/probe jobs exist) and when to run MODE=step.
- Whether carbon outputs are part of the F3 acceptance; whether to apply the binary128 Ti/Ti2b diff (speed vs bitwise).
- Whether to lift the rule 'SOCRATES is never ported' (plan: 135-295 h, decisions D1-D6).
- With the project lead: the final comparison target (published 100-year climatology vs a shorter run), identity of our reference with P2SAoM40_003, where ANN4099.aijP2SAoM40.nc lives.

Where documents disagree, and what this deck did about it

Topic	Disagreement found	Used here
Ledger range	The request mentions D1..D199; the ledger ends at D198 (HEAD ec36779).	D1-D198
Step time	GOAL.md (10-06): ~6 s per step plus clouds; D187: 44.3 s; D191: 18.8 s steady.	D191 (newest)
Remaining effort	GOAL.md ~100-170 h; README says too high, no new total; older 30-65 h understated.	no total claimed
Noise floor at steps < 3	D192/D193/D195 '~1e-9'; correction 8a649f7 shows 4e-7 (QCL, step 1).	correction
SURFACE rebuild cause	D193: tile set changes; D195: max_substeps.	D195
Day table counts	D193 Q 43/11/0, QCL 49/4/1; D195 Q 41/13/0, QCL 47/6/1.	D195
C1 array count	D187 559/559; D191 558/558 (p4_ij host index array is reference-only).	558 for D191
Summary page age	Project_Summary_and_Conclusions.md stops at D51 (680 tests); STATUS.md is Track A.	ledger + README
GPU host	ACCEPTANCE lists the GPU host as to be named; README/DISCOVER_RUN record an A100 smoke test and probe on Discover.	both stated
Jit executions in the probe note	DISCOVER_RUN says 'about 125'; D187 262, D191 99.	D187/D191

Not claimed anywhere in this deck: a pass of the day criterion; end-to-end JAX; any GPU speed-up of the full port; reproduction of the published 100-year climatology (ACCEPTANCE s5); any C2 statement from a C1 result.

Track A history (superseded) and where everything lives

Track A (2026-08-28 to 09-24), reduced scope

- Chained GHY/ATURB/SEAICE/LAKES/PBL/SURFACE driver, JAX-vectorized; measured CPU speedups 20-104x per component (Project_Summary s4A); kernels on GPU 2.3x (DRYCNV) and 6.3x (PBL); fused chain 4.2x GPU vs CPU on an A100 (STATUS.md).
- Deliberately NOT full fidelity: SEAICE, LAKES and part of ATURB were placeholders; GHY found to be one on 2026-09-23. DRYCNV is not in the real P2SAoM40 binary (ATURB is; commit b2e0487).
- The earlier deck (status_deck_2026-10-01.pdf, built by build_pdf_2026-10-01.py) reported 14 of 17 modules, 2.3-6.3x GPU kernels and a stop-the-port recommendation; the owner reversed that, so those figures are history, not current status.
- Live artifact of 2026-09-24 (deck.json, slides/*.html) is unchanged and NOT refreshed.

Sources and rebuild

- Ledger: FULL_FIDELITY_DELTAS.md (D1-D198). Index: Reports/README_START_HERE.md. Goal and status: Reports/GOAL.md. Criteria: Reports/ACCEPTANCE_CRITERIA.md (s1, 3, 4, 8, 9).
- Design: JAX_COVERAGE_MATRIX.md (stages S0-S8). Plan/provenance: RECONCILIATION_PLAN.md, PROVENANCE_MANIFEST.md. Contingency: SOCRATES_PORT_PLAN.md. Lessons: LESSONS_LEARNED.md. Chronology table: Reports/PROJECT_CHRONOLOGY.md.
- Rebuild: status_slides/README.md. In short: conda python make_figures.py DIR (matplotlib); python3 build_pdf.py status_deck.pdf --png DIR (system python3, reportlab). Core 11, scratch outside the repository.
- Stage map S0-S8 (JAX_COVERAGE_MATRIX s6): S0 harness, S1 state pytree, S2 phase 1, S3 radiation hand-off, S4 surface tiles/land, S5 post-tile + ocean, S6 assembly, S7 one coupled step, S8 multi-day.
- Review by: when the day criterion status changes, a GPU run exists, or the owner records a relaxation.